

Long Reads & T2T Genomes

Michael Schatz

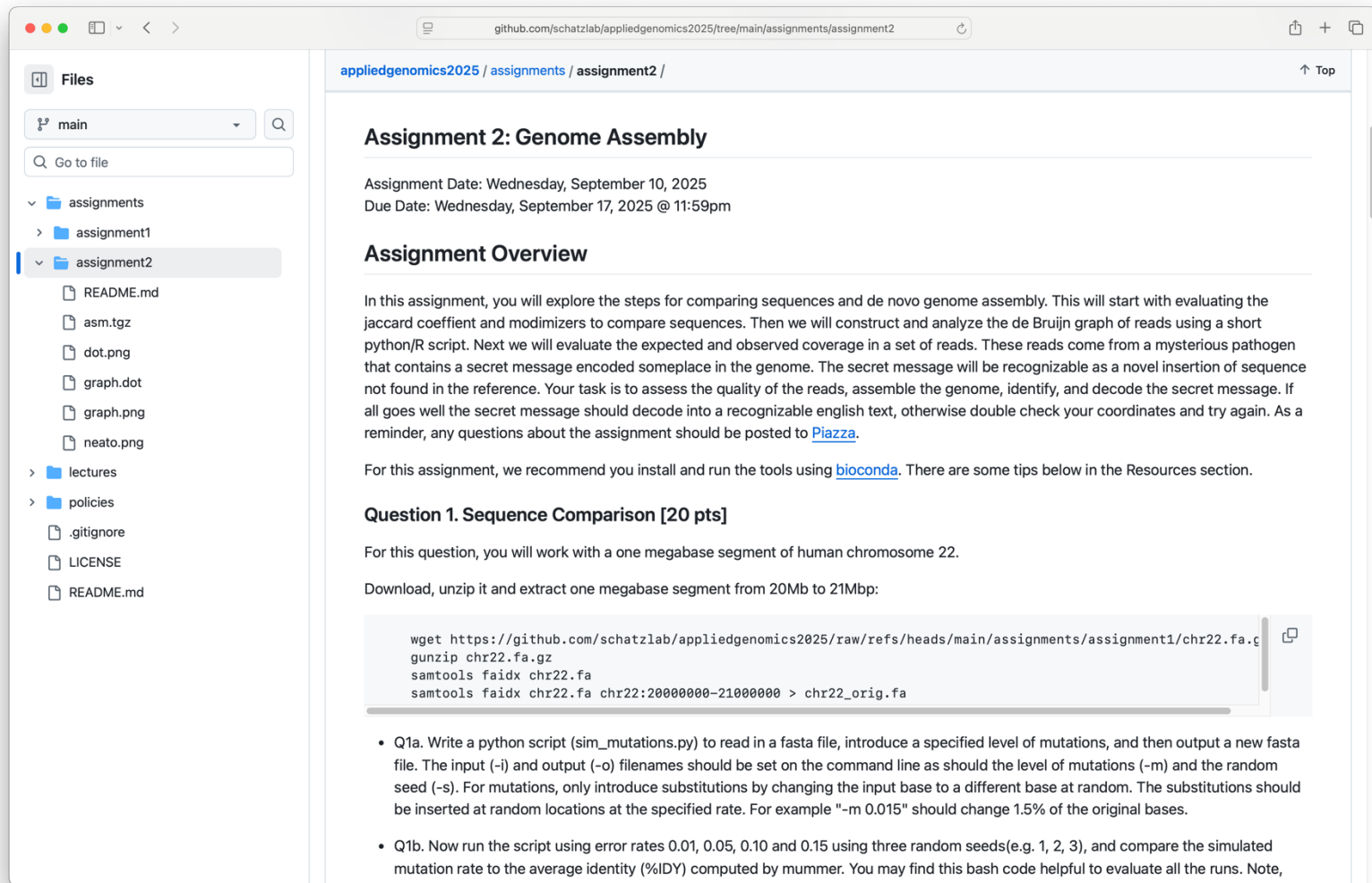
Sept 15, 2025

Lecture 6: Applied Comparative Genomics



Assignment 2: Genome Assembly

Due Wed Sept 17 by 11:59pm



Files

main

Go to file

- assignments
- assignment1
- assignment2
 - README.md
 - asm.tgz
 - dot.png
 - graph.dot
 - graph.png
 - neato.png
- lectures
- policies
- .gitignore
- LICENSE
- README.md

appliedgenomics2025 / assignments / assignment2 /

Assignment 2: Genome Assembly

Assignment Date: Wednesday, September 10, 2025
Due Date: Wednesday, September 17, 2025 @ 11:59pm

Assignment Overview

In this assignment, you will explore the steps for comparing sequences and de novo genome assembly. This will start with evaluating the jaccard coefficient and modimizers to compare sequences. Then we will construct and analyze the de Bruijn graph of reads using a short python/R script. Next we will evaluate the expected and observed coverage in a set of reads. These reads come from a mysterious pathogen that contains a secret message encoded someplace in the genome. The secret message will be recognizable as a novel insertion of sequence not found in the reference. Your task is to assess the quality of the reads, assemble the genome, identify, and decode the secret message. If all goes well the secret message should decode into a recognizable english text, otherwise double check your coordinates and try again. As a reminder, any questions about the assignment should be posted to [Piazza](#).

For this assignment, we recommend you install and run the tools using [bioconda](#). There are some tips below in the Resources section.

Question 1. Sequence Comparison [20 pts]

For this question, you will work with a one megabase segment of human chromosome 22.

Download, unzip it and extract one megabase segment from 20Mb to 21Mbp:

```
wget https://github.com/schatzlab/appliedgenomics2025/raw/refs/heads/main/assignments/assignment1/chr22.fa.gz
gunzip chr22.fa.gz
samtools faidx chr22.fa
samtools faidx chr22.fa chr22:20000000-21000000 > chr22_orig.fa
```

- Q1a. Write a python script (sim_mutations.py) to read in a fasta file, introduce a specified level of mutations, and then output a new fasta file. The input (-i) and output (-o) filenames should be set on the command line as should the level of mutations (-m) and the random seed (-s). For mutations, only introduce substitutions by changing the input base to a different base at random. The substitutions should be inserted at random locations at the specified rate. For example "-m 0.015" should change 1.5% of the original bases.
- Q1b. Now run the script using error rates 0.01, 0.05, 0.10 and 0.15 using three random seeds (e.g. 1, 2, 3), and compare the simulated mutation rate to the average identity (%IDY) computed by mummer. You may find this bash code helpful to evaluate all the runs. Note,

<https://github.com/schatzlab/appliedgenomics2025/tree/main/assignments/assignment2>

Check Piazza for questions!



The Sequence of the Human Genome

Venter et al.

Science 291, pp 1304-1351 (2001)



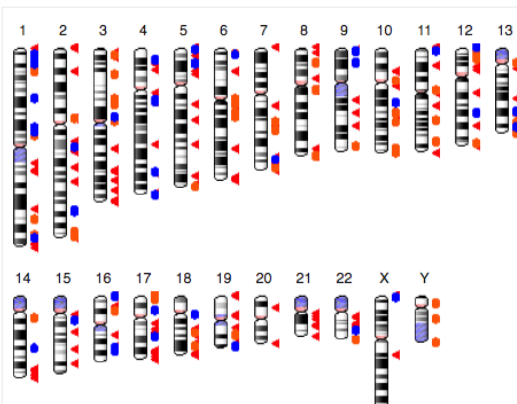
Initial sequencing and analysis of the human genome

International Human Genome Sequencing Consortium

Nature 409, pp 860-921 (2001)

Human Genome Overview

Information about the continuing improvement of the human genome



- ◀ Region containing alternate loci
- ◀ Region containing fix patches
- ◀ Region containing novel patches

Ideogram of the latest human assembly, GRCh38.p11

[GRCh38.p11](#)
[GRCh37.p13](#)
[GRCh37](#)

GRCh38.p11

Release date: June 14, 2017

Release type: minor

Release notes: GRCh38.p11 is the eleventh patch release for the GRCh38 reference assembly. No chromosome coordinate changes. The number of patch scaffolds is now: 64 FIX and 59 NOVEL.

Assembly accessions: GenBank: [GCA_000001405.26](#) , RefSeq: [GCF_000001405.37](#)

Pseudoautosomal regions

Name	Chr	Start	Stop
PAR#1	X	10,001	2,781,479
PAR#2	X	155,701,383	156,030,895
PAR#1	Y	10,001	2,781,479
PAR#2	Y	56,887,903	57,217,415

The GRC is working hard to provide the best possible reference by both generating multiple representations (alternates) and by representing a single path. Additionally, we are releasing alternate representations that allows users who are interested in a specific locus to view the data without affecting users who need chromosome coordinate space.

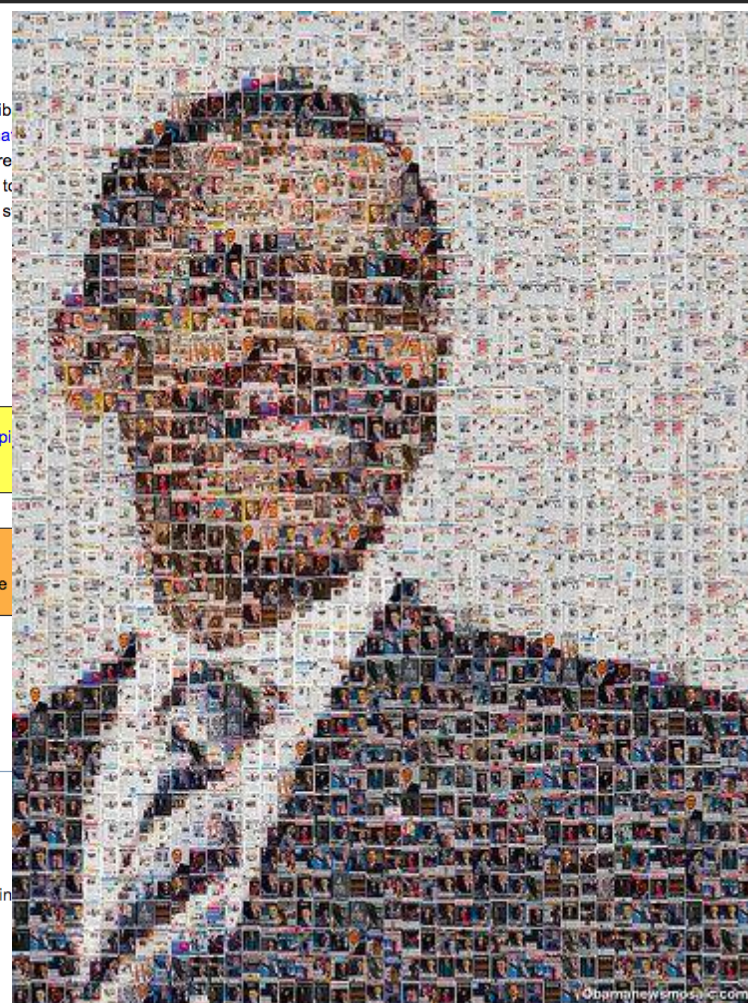
Download data:

- [GRCh38.p11 \(latest minor release\) FTP](#)
- [GRCh38 \(latest major release\) FTP](#)
- [Genomic regions under review FTP](#)
- [Current Tiling Path Files \(TPFs\)](#)

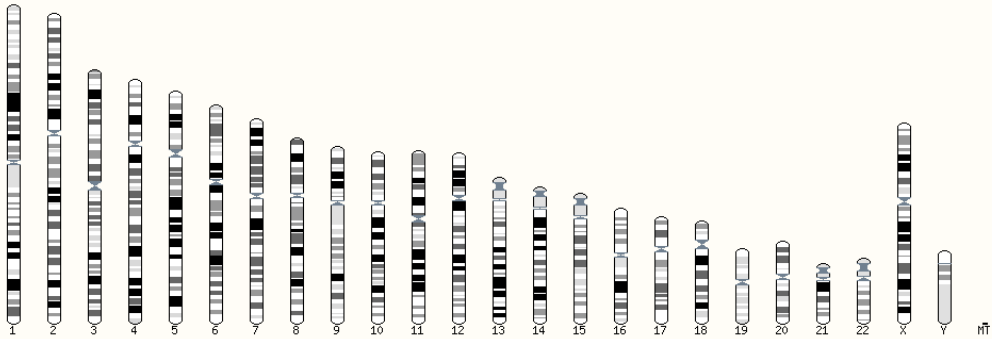
Transitioning to GRCh38? Try the [NCBI Remapping tool](#) for assembly alignments used by the GRC.

Next assembly update

The next assembly update (GRCh38.p12) will be released in the near future.



The human genome - basic stats

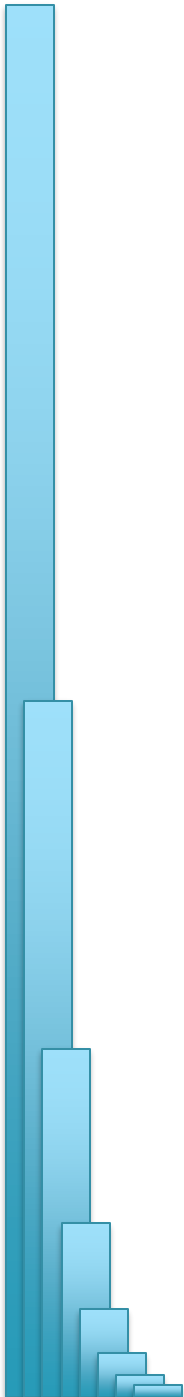


- 3.096 billion base pairs (haploid)
- 20,454 protein coding genes
- 226,950 coding transcripts
(isoforms of a gene that each encode a distinct protein product)

Assembly	GRCh38.p12 (Genome Reference Consortium Human Build 38), INSDC Assembly GCA_000001405.27 , Dec 2013
Base Pairs	3,609,003,417
Golden Path Length	3,096,649,726
Annotation provider	Ensembl
Annotation method	Full genebuild
Genebuild started	Jan 2014
Genebuild released	Jul 2014
Genebuild last updated/patched	Mar 2019
Database version	97.38
Gencode version	GENCODE 31

Gene counts (Primary assembly)

Coding genes	20,454 (incl 660 readthrough)
Non coding genes	23,940
Small non coding genes	4,871
Long non coding genes	16,848 (incl 302 readthrough)
Misc non coding genes	2,221
Pseudogenes	15,204 (incl 8 readthrough)
Gene transcripts	226,950

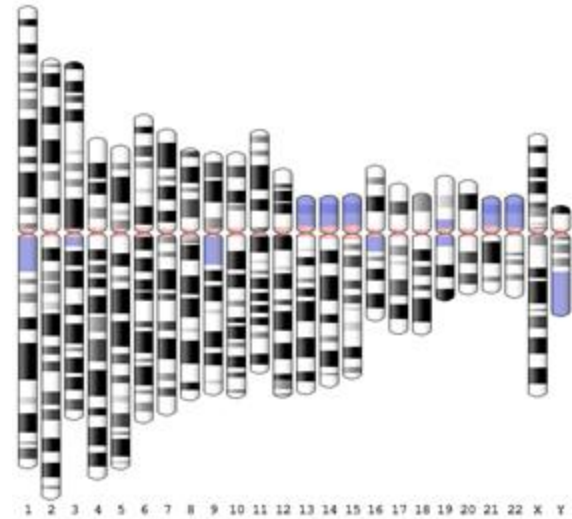


The T2T human genome

Finishing the human genome

238Mbp is missing or incorrect

- Centromeres and telomeres
- Segmentally duplicated genes
- Tandem gene arrays (e.g. rDNAs)
- And an unknown number of errors...



Why does it matter?

Variation in these regions is unexplored

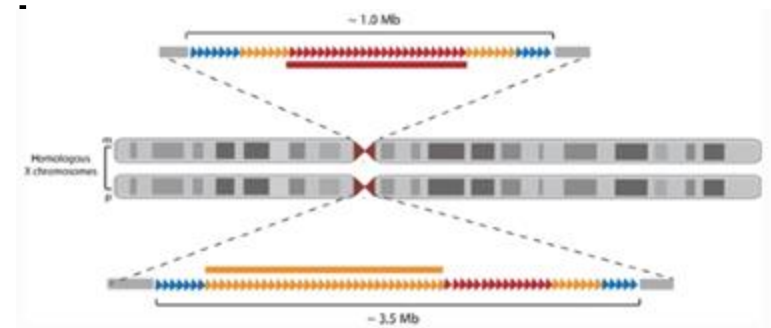
Functional studies need sequence

Reference gaps lead to artifacts

We don't know what we don't

Why has it taken so long?

Repeats, repeats, repeats...



Let's finish a human genome



T2T Working Group

[Home](#)

[Technology](#)

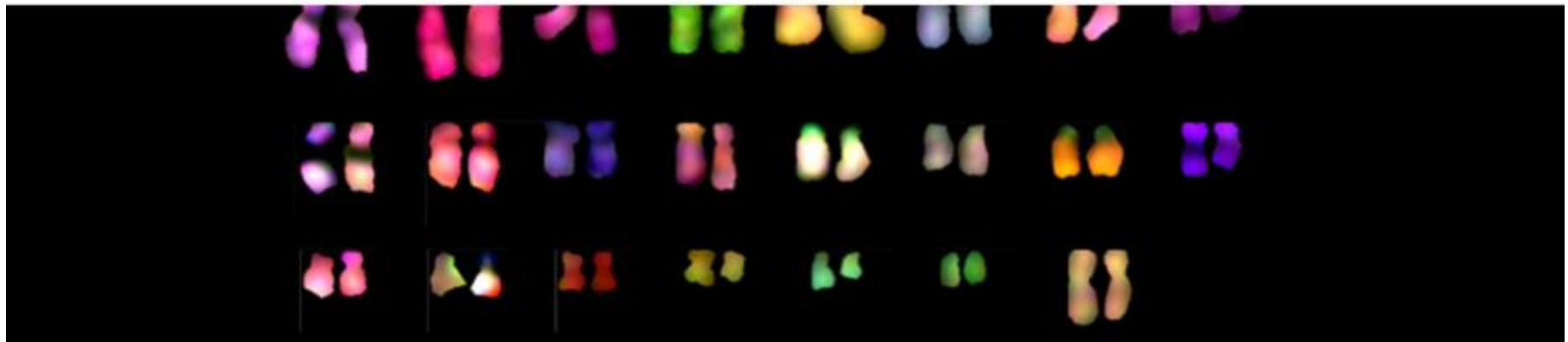
[Data](#)

[CHM13 Cell Line](#)

[Remaining Challenges](#) ▼

[Who We Are](#)

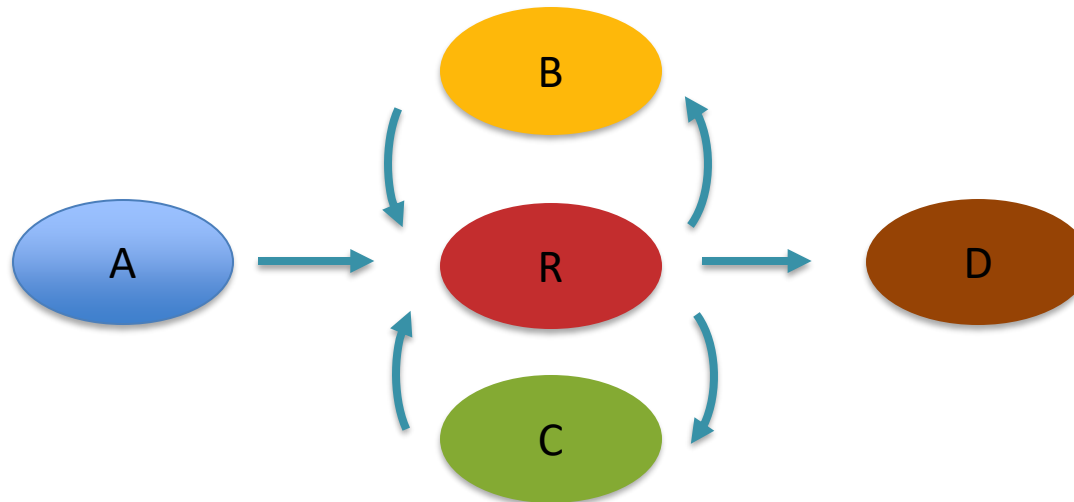
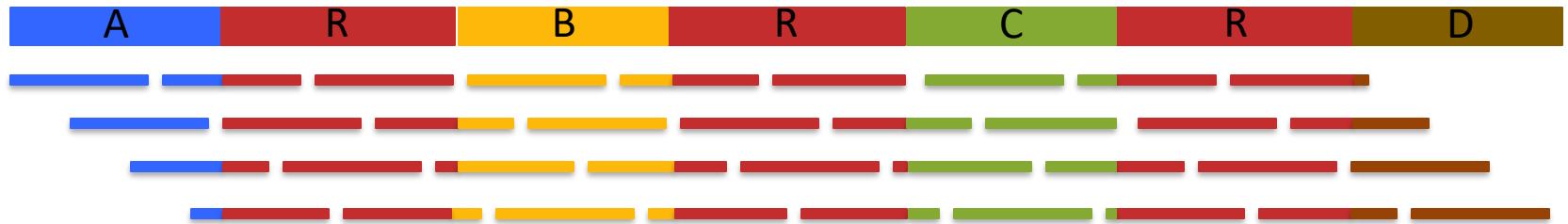
[Join Us](#)



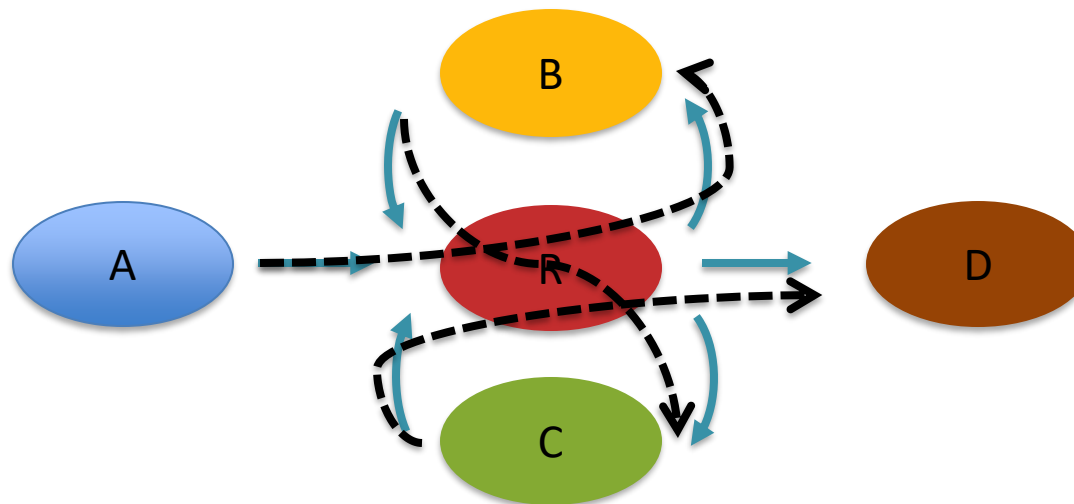
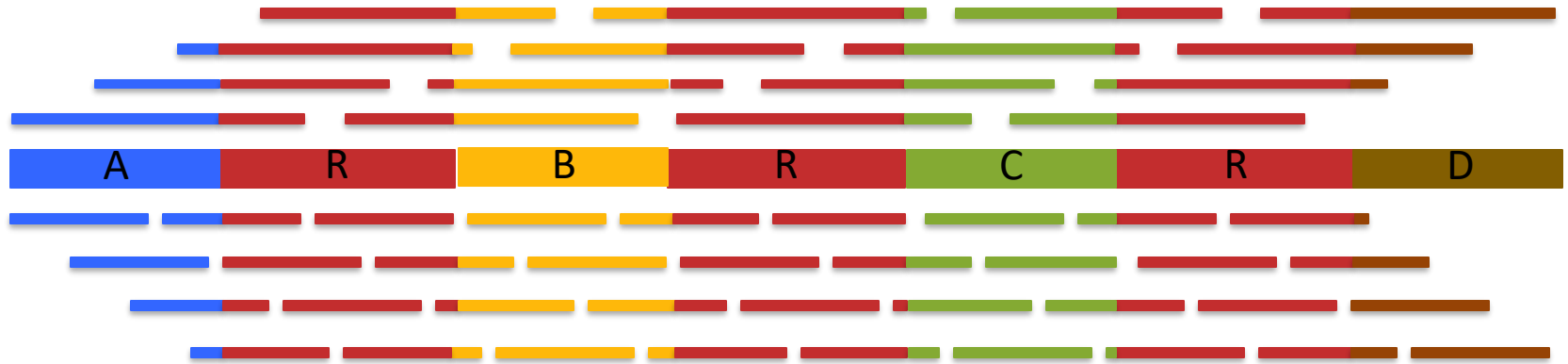
The Telomere-to-Telomere (T2T) consortium is an open, community-based effort to generate the first complete assembly of a human genome.

CHM13 homozygous 46,XX cell line from Urvashi Surti, Pitt; SKY karyotype from Jennifer Gerton, Stowers

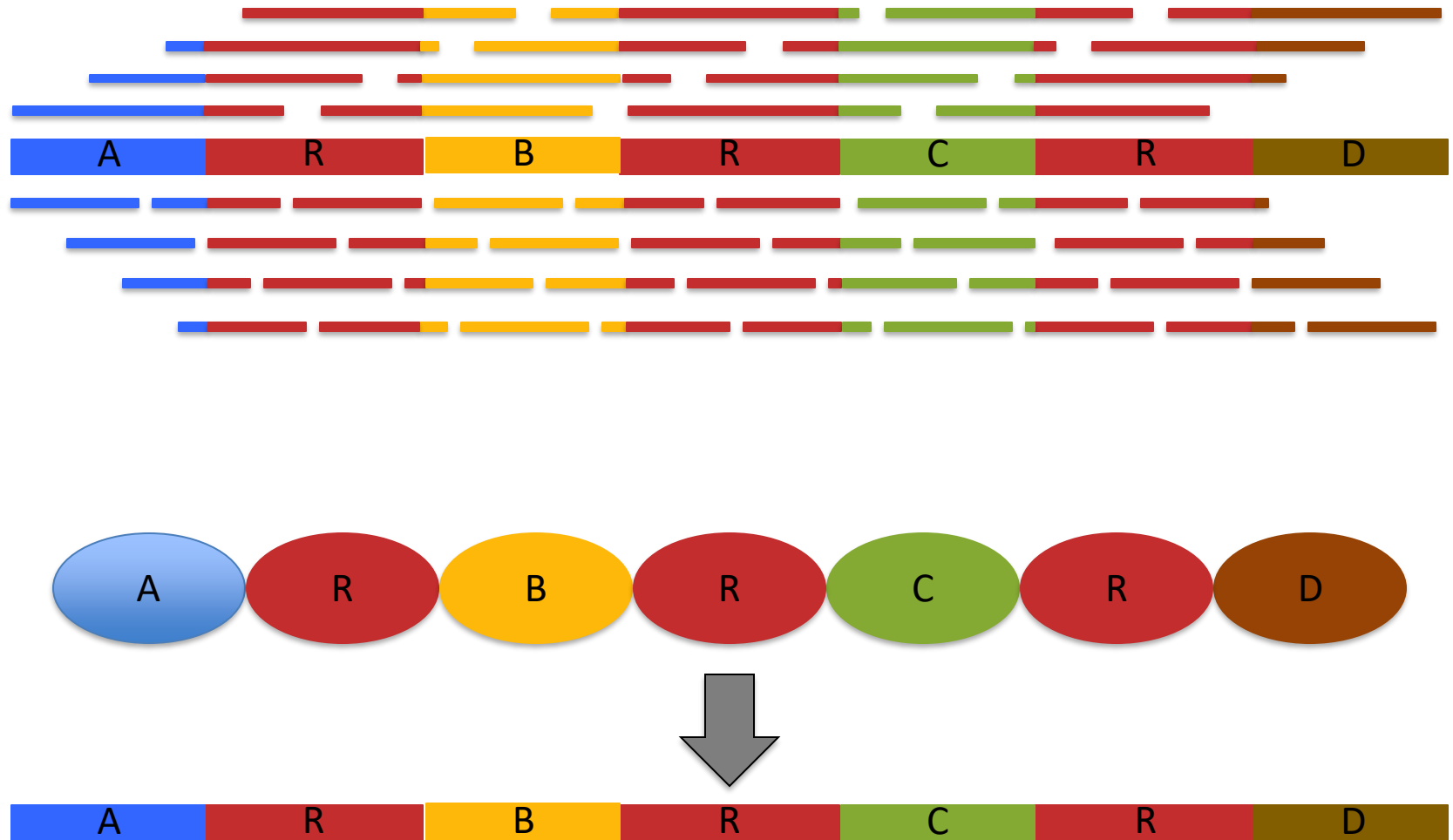
Assembly Complexity



Assembly Complexity



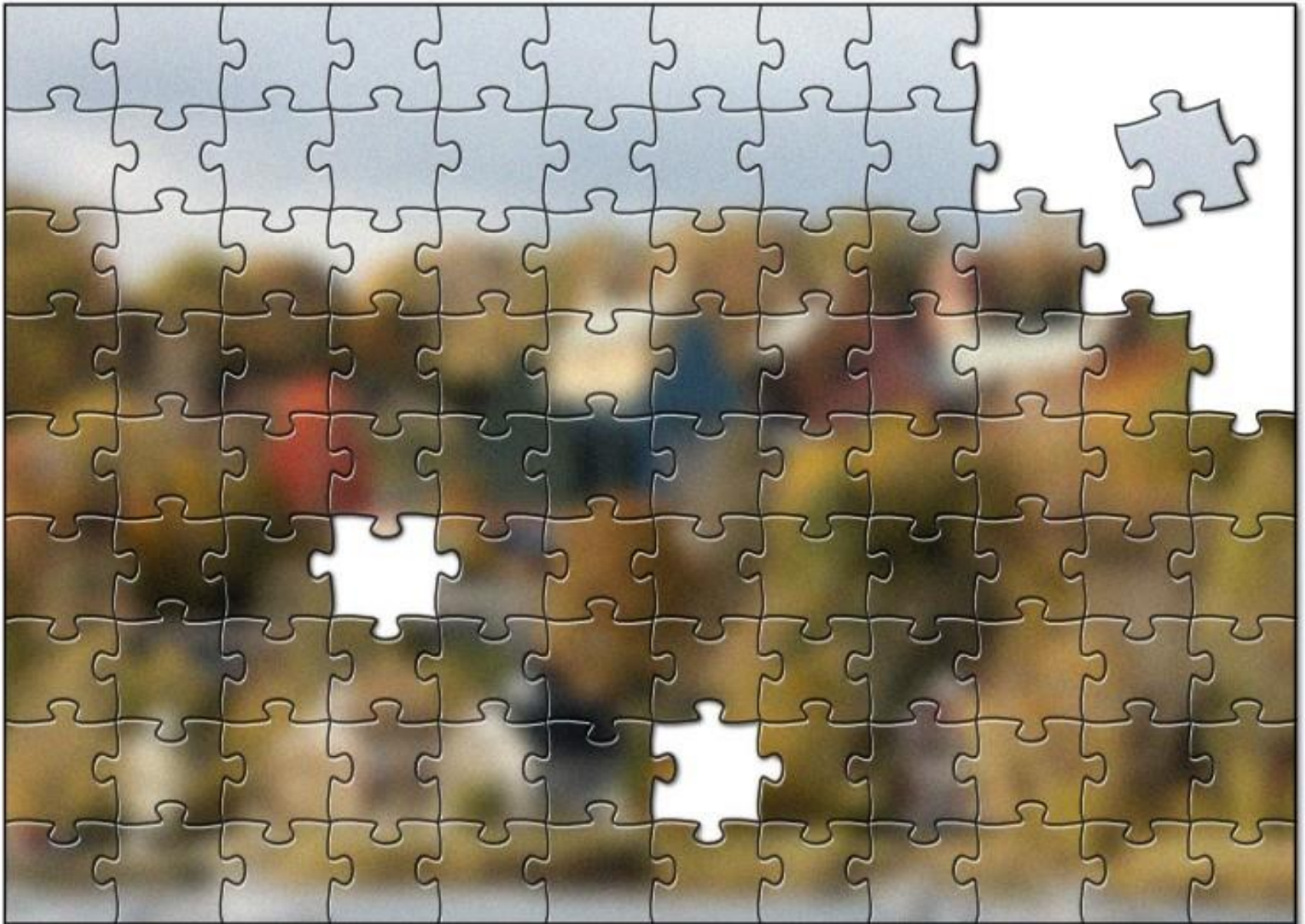
Assembly Complexity



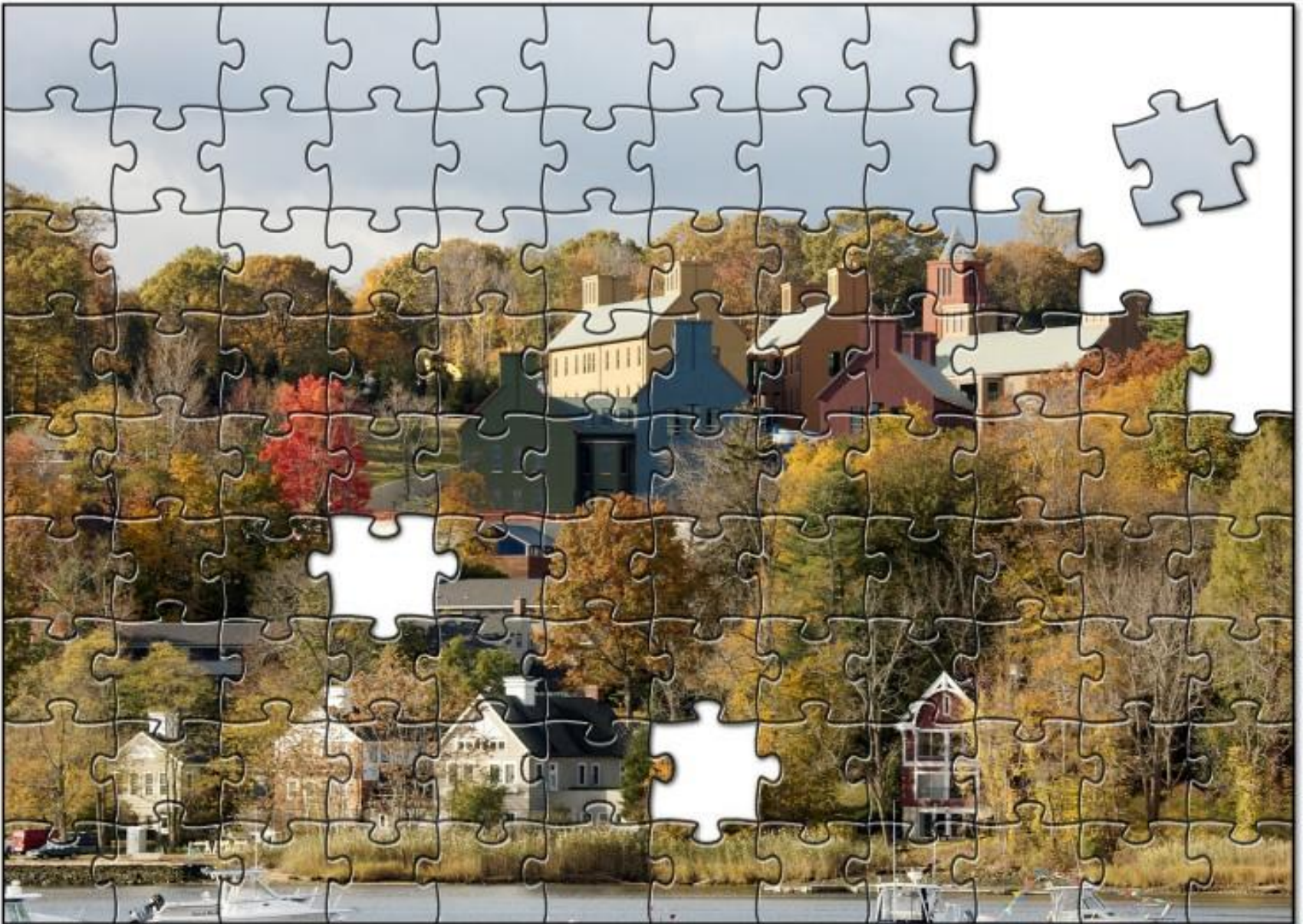
The advantages of SMRT sequencing

Roberts, RJ, Carneiro, MO, Schatz, MC (2013) *Genome Biology*. 14:405

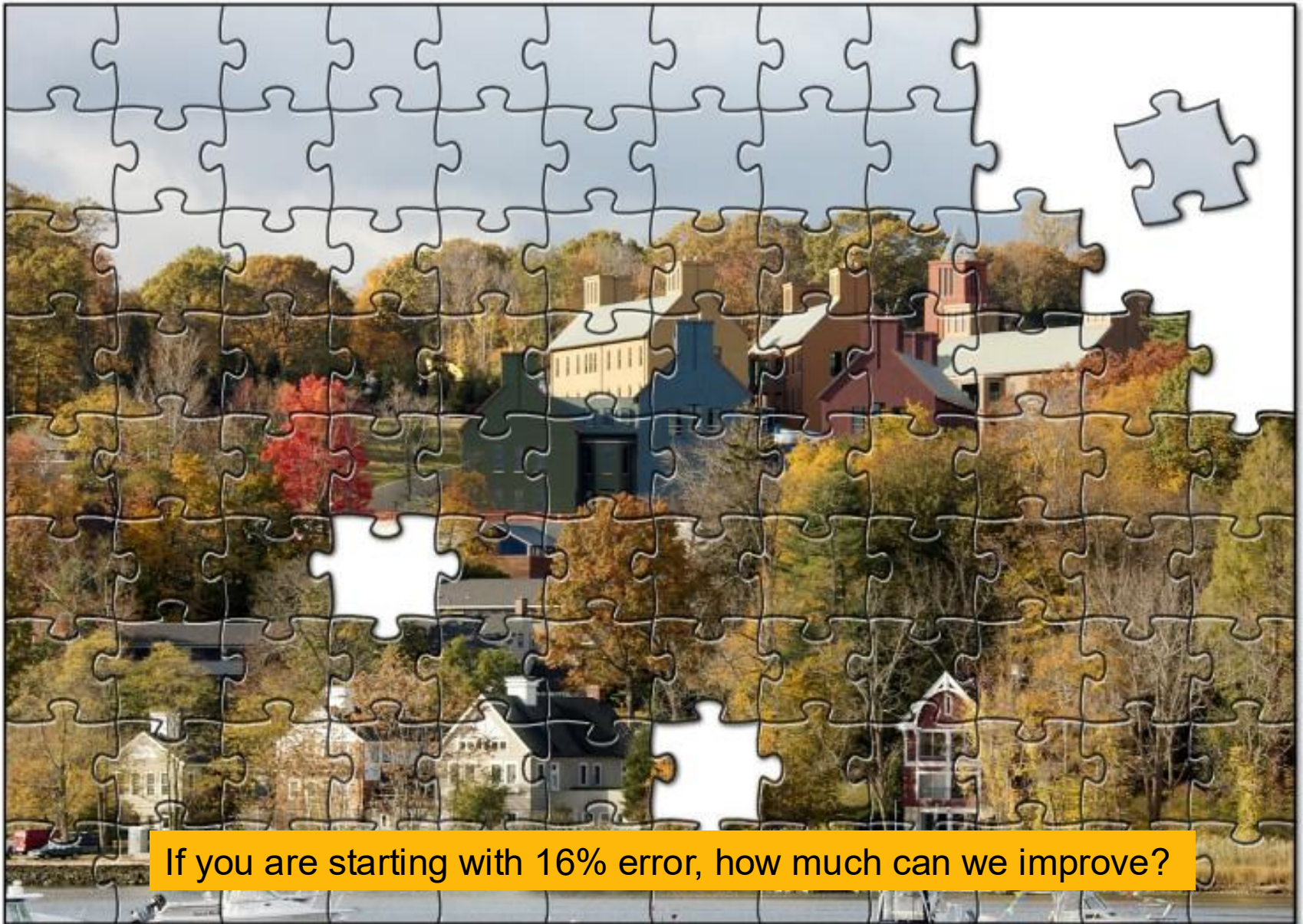
Single Molecule Sequences



“Corrective Lens” for Sequencing

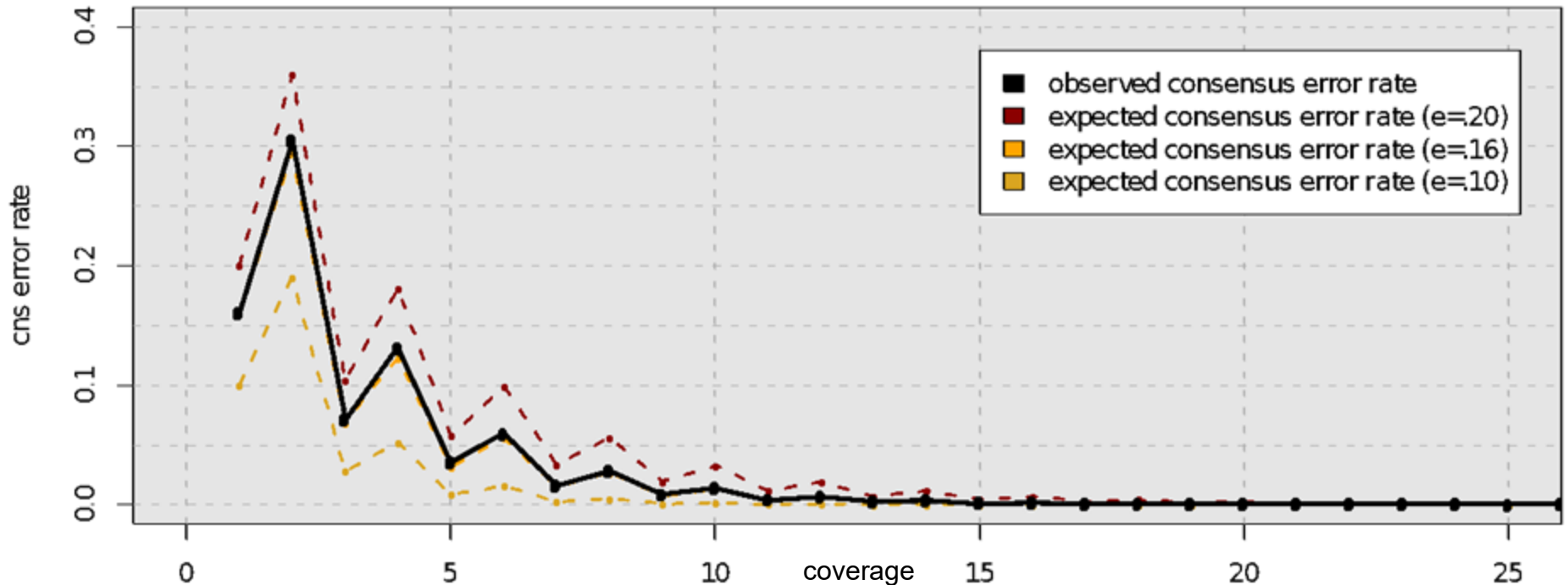


“Corrective Lens” for Sequencing



If you are starting with 16% error, how much can we improve?

Consensus Accuracy and Coverage

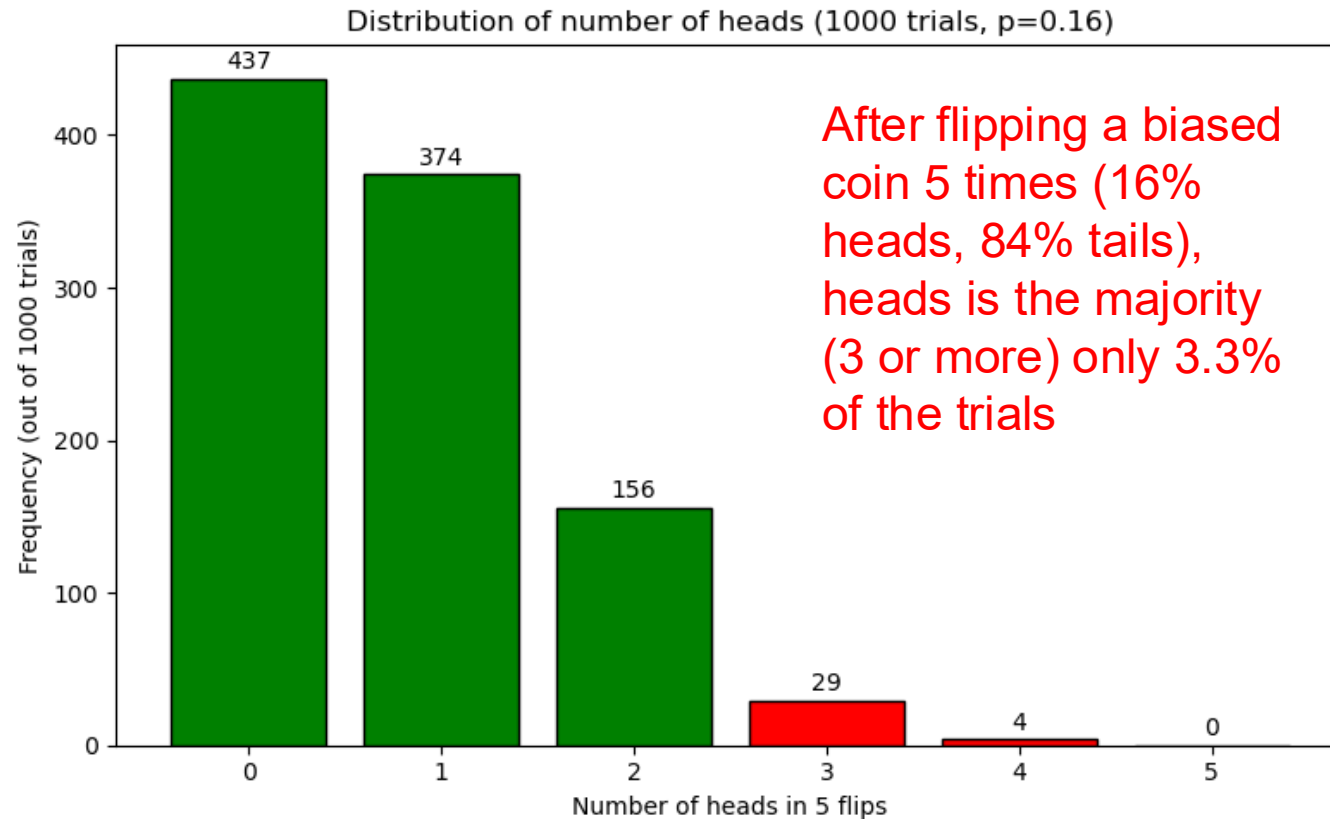


Coverage can overcome random errors

- Dashed: error model from binomial sampling; solid: observed accuracy
- For same reason, CCS is extremely accurate when using 5+ subreads

$$CNS\ Error = \sum_{i=\lceil c/2 \rceil}^c \binom{c}{i} (e)^i (1-e)^{n-i}$$

Consensus Accuracy and Coverage



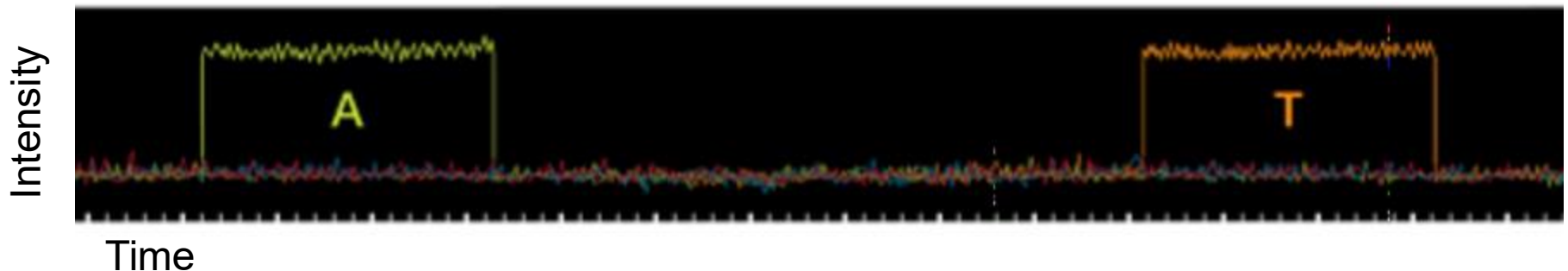
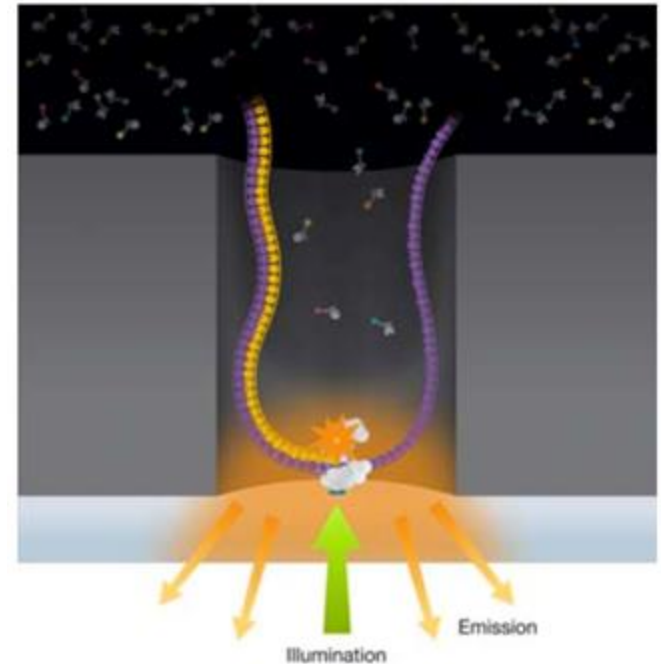
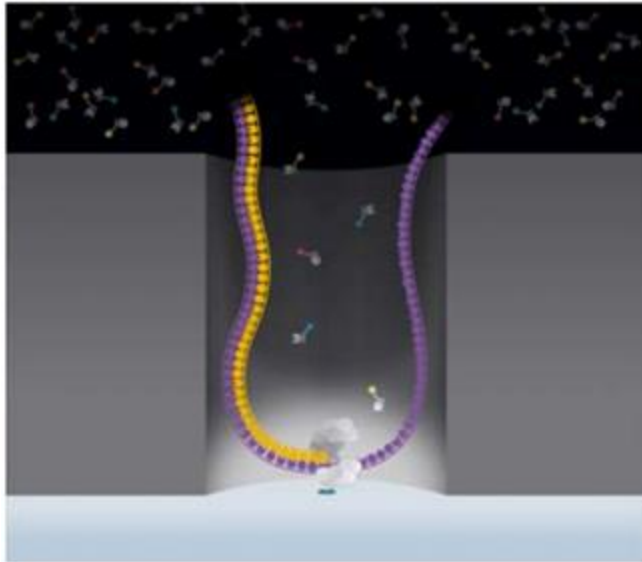
GPT5-mini: “Write python code to simulate 1000 trials of flipping a biased coin 5 times where there is a 16% chance of heads and 84% chance of tails. Then plot a histogram of the distribution of the number of heads. colors the bars green when there < 3 heads and red if there are 3 or more”

PacBio: SMRT Sequencing



PacBio: SMRT Sequencing

Imaging of fluorescent phospholinked labeled nucleotides as they are incorporated by a polymerase anchored to a Zero-Mode Waveguide (ZMW).



“HiFi” Circular Consensus Reads

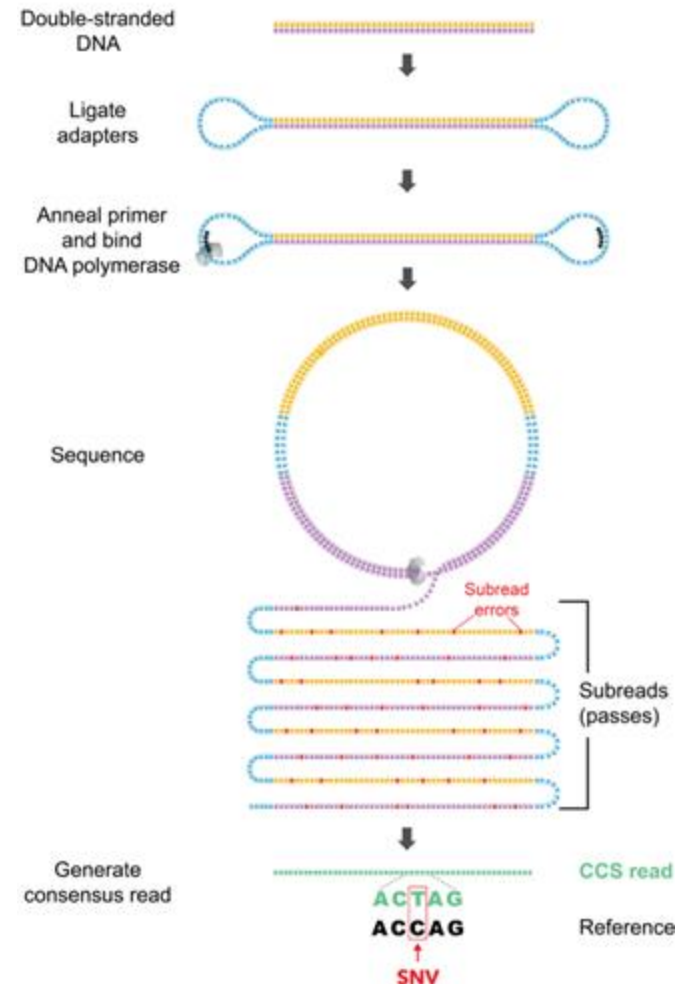
High-quality reads produced by sequencing the same molecule multiple times

Higher accuracy for low-coverage sequences like somatic variants or lowly expressed transcripts in RNA-seq, more interpretable alignments, better & faster assembly

Limits read length, used to be very expensive but more manageable now

Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome

Wenger et al (2019) Nature Biotechnology doi:10.1038/s41587-019-0217-9

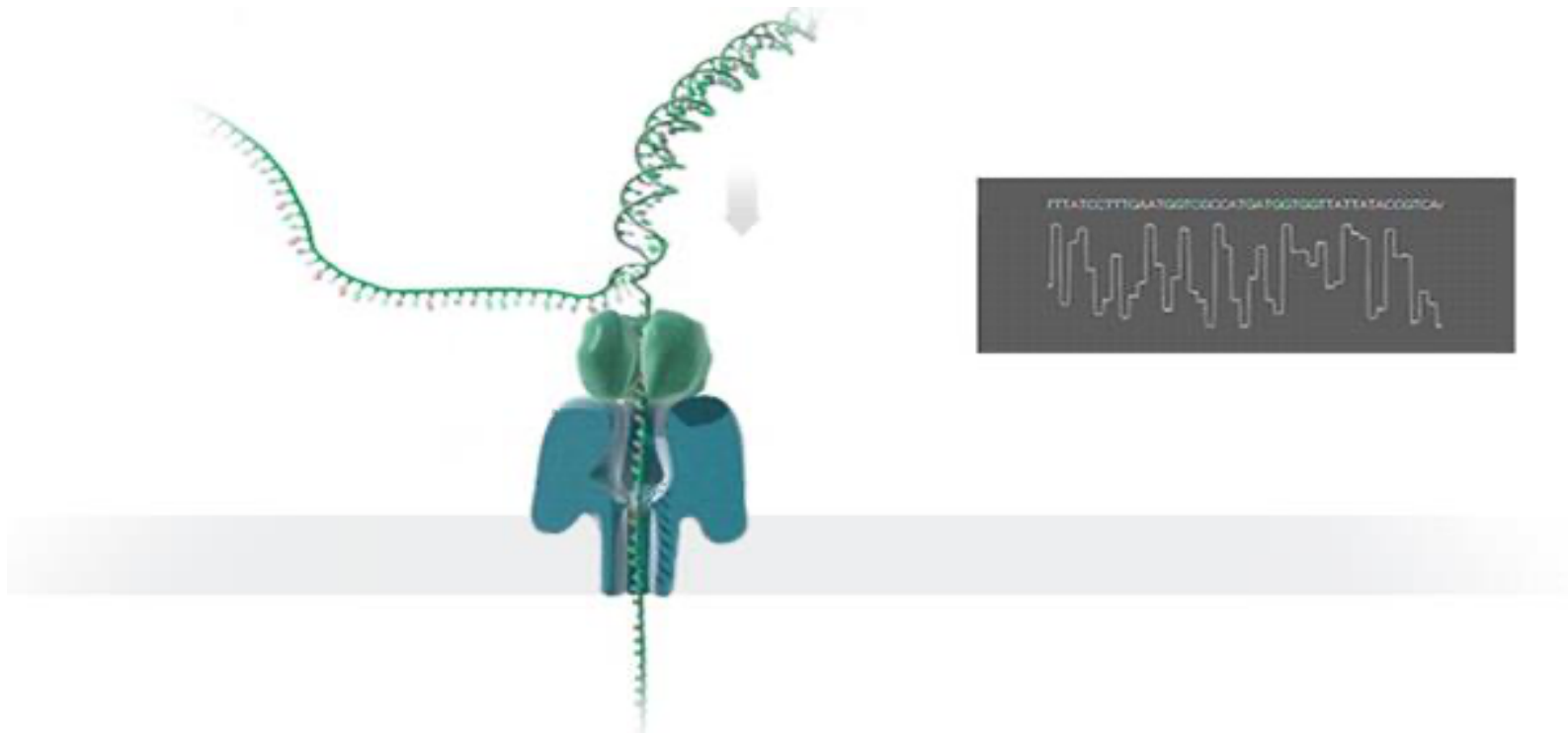


Oxford Nanopore Technologies (ONT)



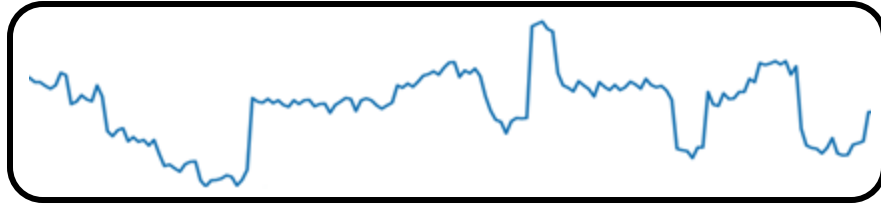
Nanopore Sequencing

Sequences DNA/RNA by measuring changes in ionic current as nucleotide strand passes through a pore



Nanopore Basecalling

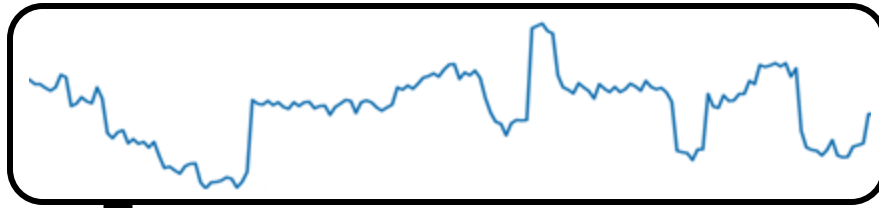
Raw Signal



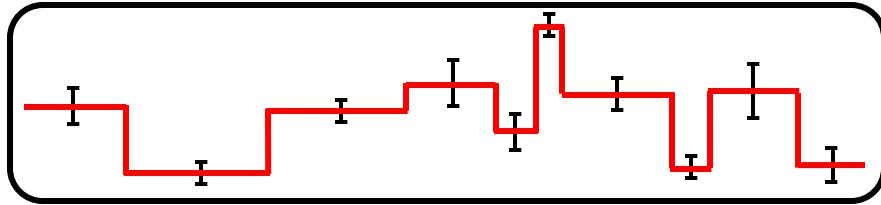
Translation of raw signal
into basepairs

Nanopore Basecalling

Raw Signal



Events



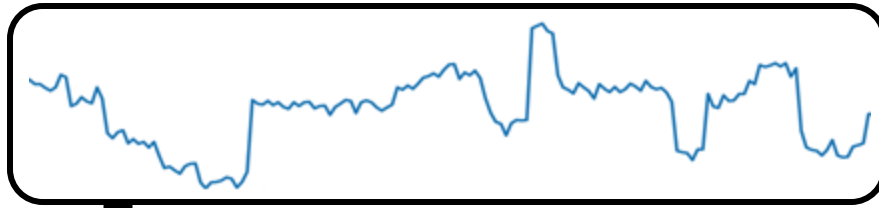
Translation of raw signal into basepairs

Early basecallers began by estimating k-mer boundaries using “events”, which were then input to an HMM

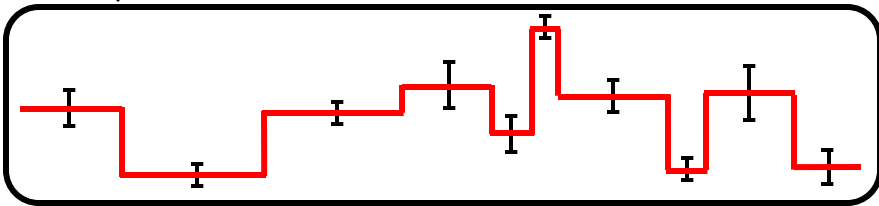
Modern basecallers use neural networks directly on raw signal

Nanopore Basecalling

Raw Signal



Events

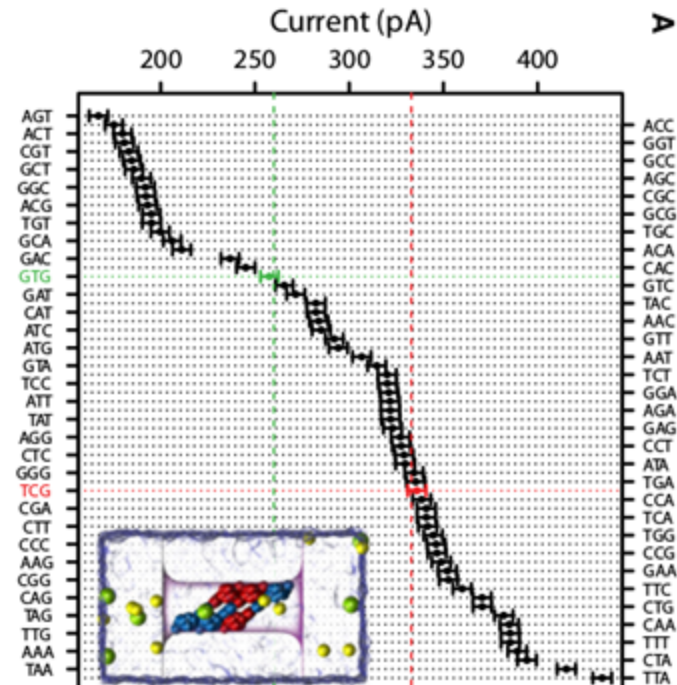


Possible k-mers

0	1	2	3
TCCA	CCAT	CATG	TACA
AGCA	TGGC	TTAC	TCCA
GTCT	ATTA	ACGT	GACG
GATT	ATTG	GTCT	ACGG

(Based on probability of event matches)

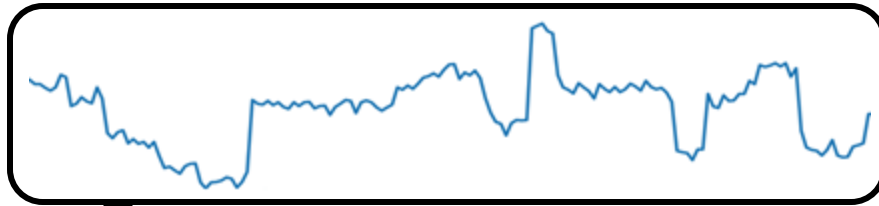
ONT releases k-mer models with expected current distribution of every k-mer



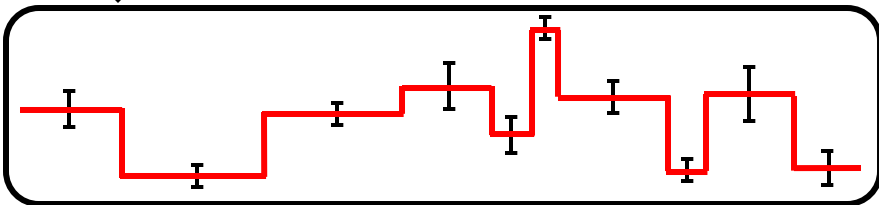
DNA Base-Calling from a Nanopore Using a Viterbi Algorithm
Timp et al. (2012) *Biophysical Journal*

Nanopore Basecalling

Raw Signal



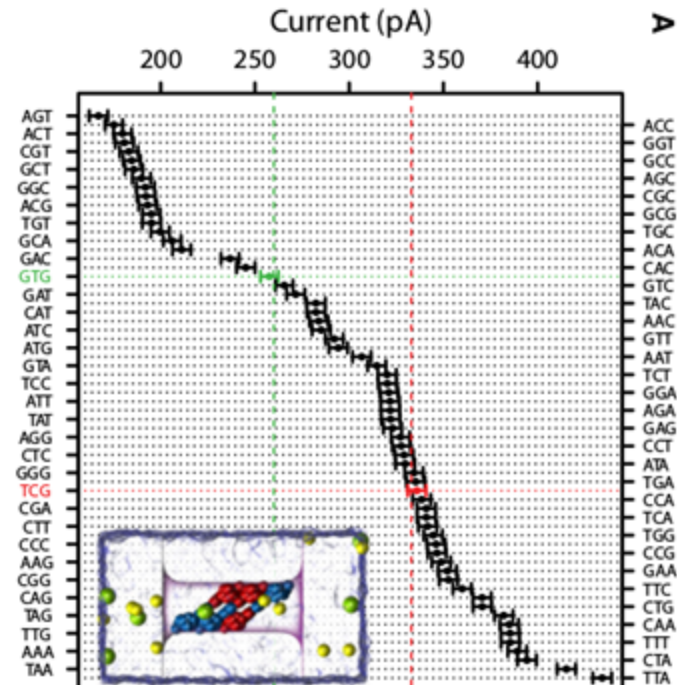
Events



Possible k-mers

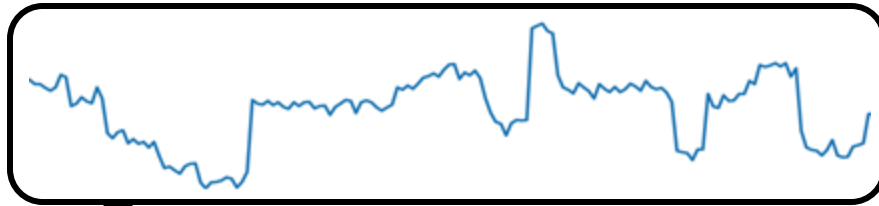
0	1	2	3
TCCA → CCAT	CATG	TACA	
AGCA → TGGC	TTAC	TCCA	
GTCT → ATTA	ACGT	GACG	
GATT → ATTG	GTCT	ACGG	

Certain k-mers can be eliminated based on possible transitions

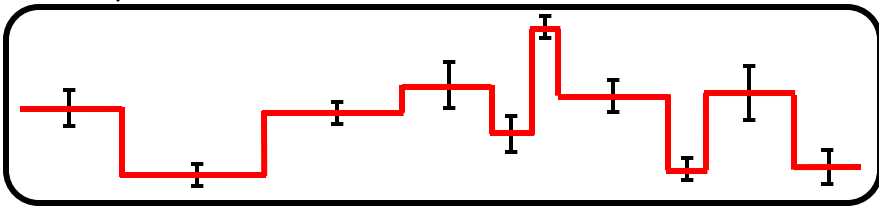


Nanopore Basecalling

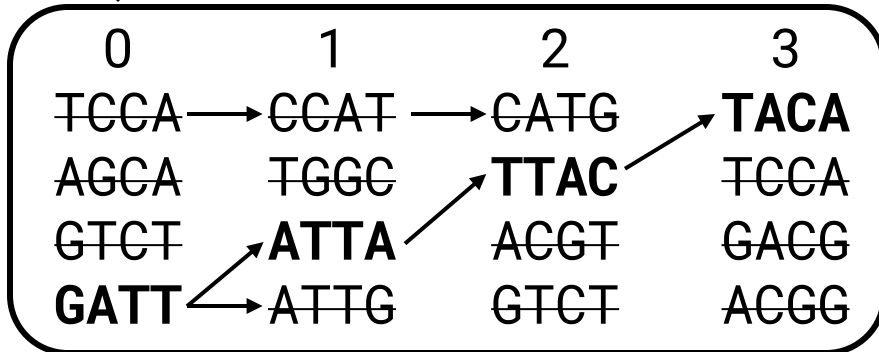
Raw Signal



Events

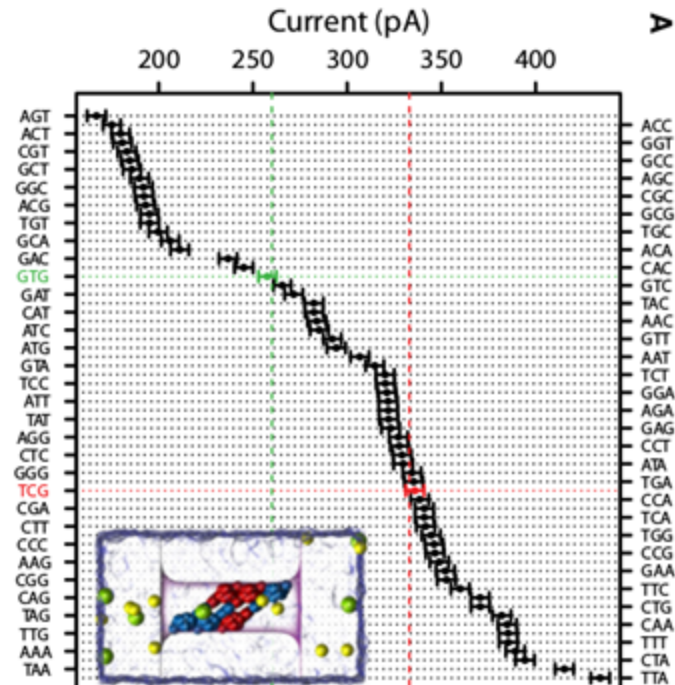


Possible k-mers



GATTACA

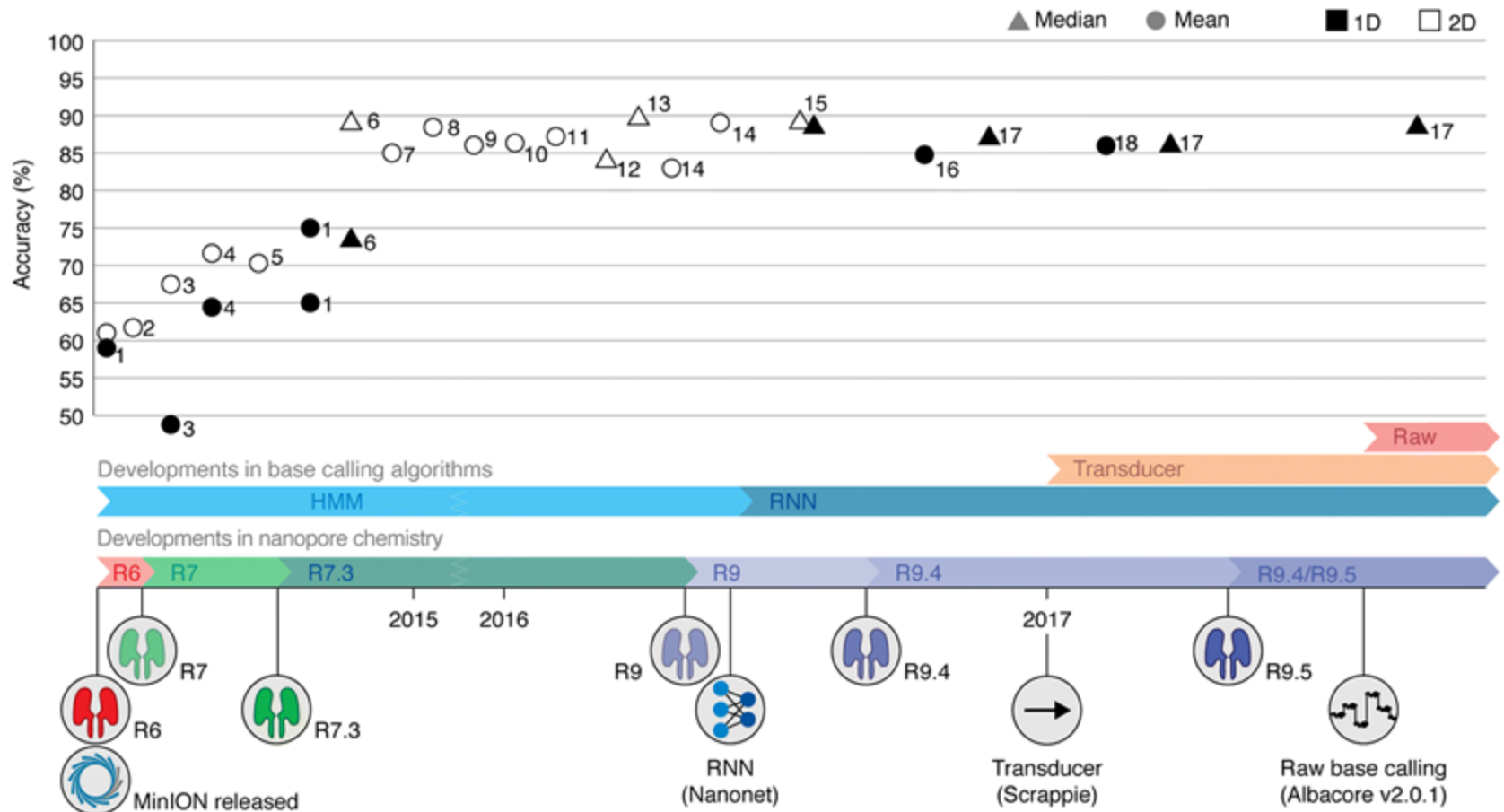
Final sequence determined by most probable k-mers



"DNA Base-Calling from a Nanopore Using a Viterbi Algorithm"
Timp et al. (2012) *Biophysical Journal*

Basecaller/Pore Timeline

Development of both pore chemistry and basecalling algorithms is responsible for improvement in accuracy

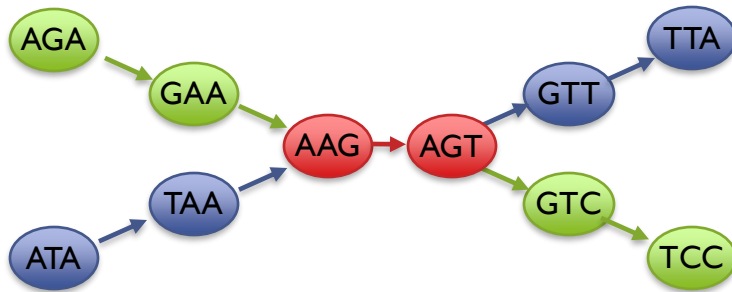


From squiggle to basepair: computational approaches for improving nanopore sequencing read accuracy

Rang et al (2018) *Genome Biology*. <https://doi.org/10.1186/s13059-018-1462-9>

Two Paradigms for Assembly

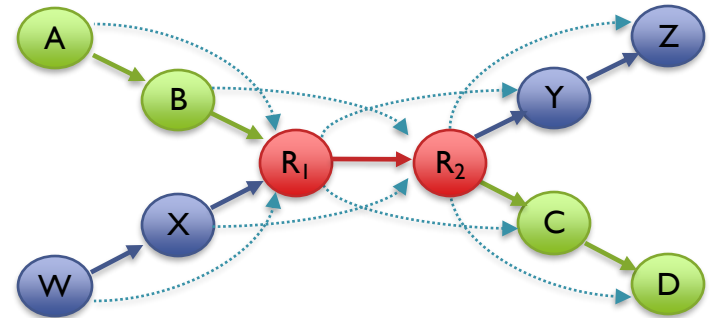
de Bruijn Graph



Short read assemblers

- Repeats depends on word length
- Read coherency, placements lost
- Robust to high coverage

Overlap Graph



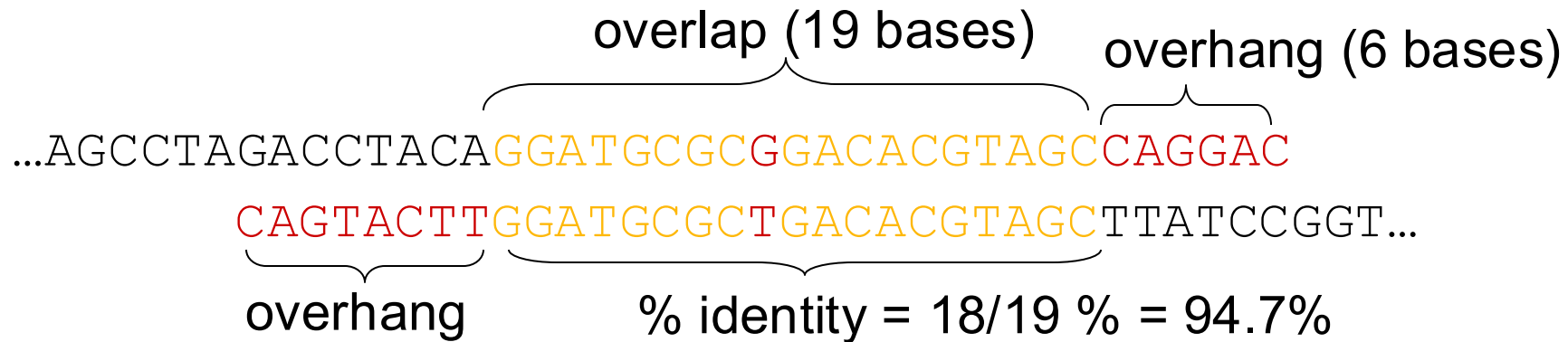
Long read assemblers

- Repeats depends on read length
- Read coherency, placements kept
- Tangled by high coverage

Assembly of Large Genomes using Second Generation Sequencing

Schatz MC, Delcher AL, Salzberg SL (2010) *Genome Research*. 20:1165-1173.

Overlap between two sequences



overlap - region of similarity between regions

overhang - un-aligned ends of the sequences

The assembler screens merges based on:

- length of overlap
- % identity in overlap region
- maximum overhang size.

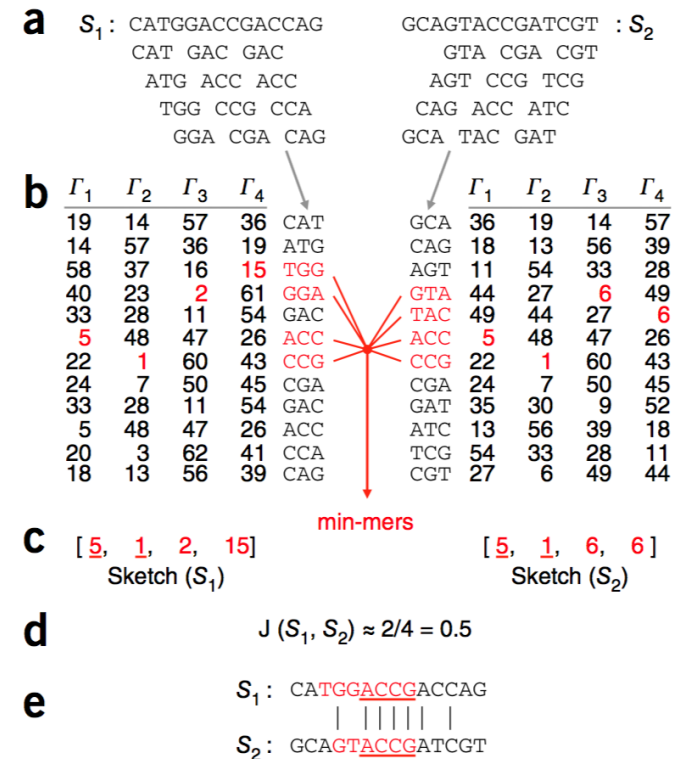
[How do we compute the overlap?]

[Do we really want to do all-vs-all?]

Very fast approximate overlapping

Maybe we don't need to compute the exact identity of the overlap region, just approximate it

- If two reads overlap, they should share many of the same kmers: Their Jaccard coefficient should be high: $|\text{intersection}| / |\text{union}|$
- But tracking all of the kmers for a read is a lot of overhead
- Instead, compare the “sketch” of the reads: a small fraction of kmers carefully chosen
- LSH: Find the sketch by applying N hash functions to the kmers, and keeping the minimum hash values reported from each (N=4 in example)
- This forms a nice “random” sample of the reads, and the Jaccard coefficient is a good approximation of the sequence similarity



Assembling large genomes with single-molecule sequencing and locality-sensitive hashing

Berlin et al (2015) *Nature Biotechnology*

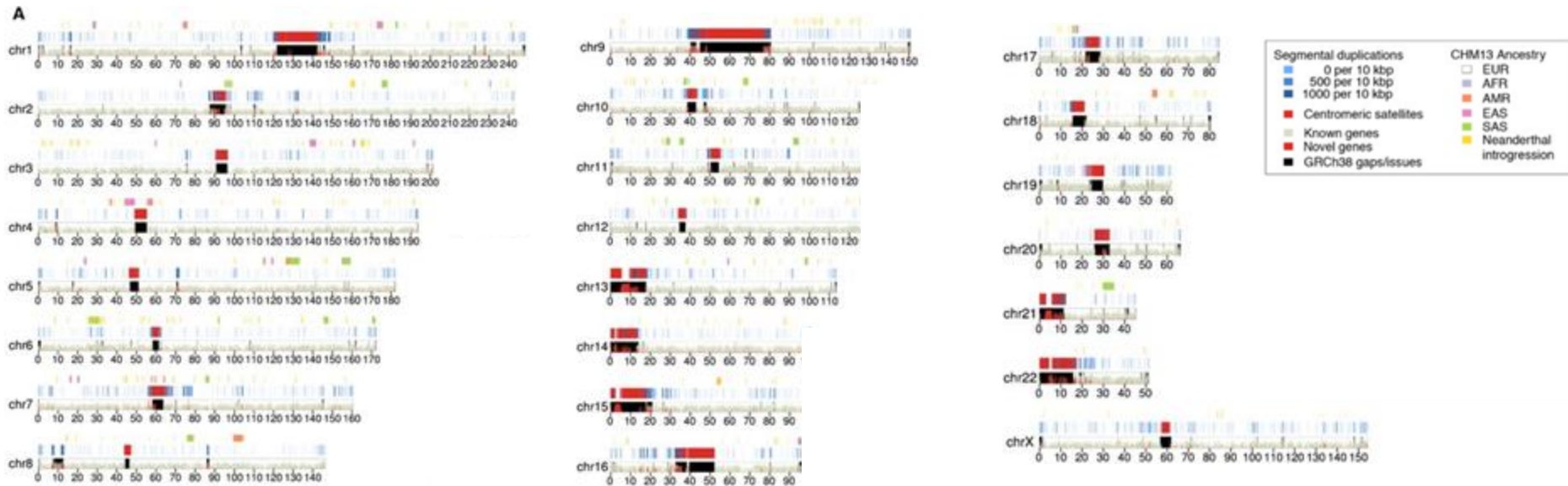
CHM13 assembly graph



HiCanu: accurate assembly of segmental duplications, satellites, and allelic variants from high-fidelity long reads.

Nurk et al. *Genome Research* (2020)

The complete sequence of a human genome



CHM13v1.1 genome size is **3.057 Gbp with zero Ns**

Every chromosome is telomere-to-telomere, quality estimated >Q70
~**190 Mbp (~8%)** of new sequence vs. GRCh38, fixes thousands of errors

(Nurk et al. Science, 2022)

Earth's heart of iron begins
to yield its secrets p. 18

Microglia in chronic pain recovery
and relapse pp. 33 & 86

Particle acceleration
in a nova explosion p. 77

Science

\$15
1 APRIL 2022
SPECIAL ISSUE
science.org

AAAS

FILLING THE GAPS

Closing in on a complete
human genome p. 42

NEWS CAREERS COMMENTARY JOURNALS

Science

LOG IN BECOME A MEMBER

HOME > COLLECTIONS > COMPLETING THE HUMAN GENOME

COMPLETING THE HUMAN GENOME

A fully sequenced human genome was announced more than 20 years ago. However, owing to technological limitations, some genomic regions remained unresolved. Here, *Science* and other journals present research by the Telomere-to-Telomere (T2T) Consortium, reporting on the endeavor to complete a comprehensive human reference genome.

6 FILTERS 6 RESULTS FOUND

SPECIAL ISSUE RESEARCH ARTICLE

Segmental duplications and their variation in a complete human genome

BY MITCHELL R. VOLLGER, RAVI GUTART, PHILIP C. DISHACK, LUDOVICA MERCURI, WILLIAM T. HARVEY, ARIEL GERSHMAN, MARK DEKHAHI, ARVID SULOVARI, KATHERINE M. MUNDON, ALEXANDRA P. LEWIS, [...] EVAN E. EICHLE

SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

Complete genomic and epigenetic maps of human centromeres

BY NICOLAS ALTEMOSE, GLENNIS A. LOGSDON, ANDREY V. BZKADEZ, PRADYU BOHRAI, SASHA A. LANGLEY, DINA V. CALDAS, SAVANNAH J. HOYT, LEV URALSKY, FEDOR D. RYABOV, COLIN J. SHEW, [...] KAREN H. NIGDA

SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

From telomere to telomere: The transcriptional and epigenetic state of human repeat elements

BY SAVANNAH J. HOYT, JESSICA M. STORER, GABRIELLE A. HARTLEY, PATRICK G. S. GRADY, ARIEL GERSHMAN, LEONARDO G. DE LIMA, CHARLES LMOUSE, REZA HALABIAN, LUKE WOLJENSKI, MATIAS RODRIGUEZ, [...] RACHEL A.

16 authors • SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

A complete reference genome improves analysis of human genetic variation

BY SERGEY AGANEZOV, STEPHANIE M. YAN, DANIELA C. SOTO, MELANIE KIRSCH, SAMANTHA ZARATE, PAVEL AVDEYEV, DYLAN J. TAYLOR, KISHAN SHAFIN, ALAINA SHUMATE, CHUNLIN XIAO, [...] MICHAEL C. SCHATZ

SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

Epigenetic patterns in a complete human genome

BY ARIEL GERSHMAN, MICHAEL C. SCHATZ, SAURIA, RAVI GUTART, MITCHELL R. VOLLGER, PAUL W. HOOK, SAVANNAH J. HOYT, METEN JIAN, ALAINA SHUMATE, ROHAM RAZAGHI, SERGEY KOREN, [...] WINSTON TAMP

9 authors • VOL. 376, NO. 6588 • 01 APR 2022

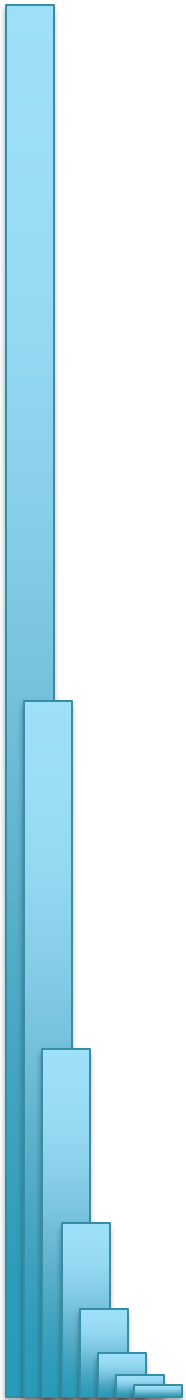
SPECIAL ISSUE RESEARCH ARTICLE

The complete sequence of a human genome

BY SERGEY KOREN, ARAND RHIE, MIKKO RAUTAIENEN, ANDREY V. BZKADEZ, ALLA MIKHENKO, MITCHELL R. VOLLGER, NICOLAS ALTEMOSE, LEV URALSKY, ARIEL GERSHMAN, [...] ADAM M. PHILLIPPY

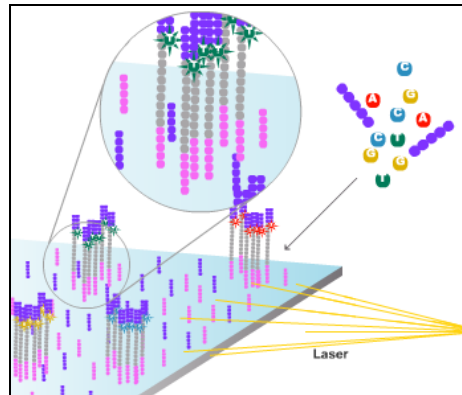
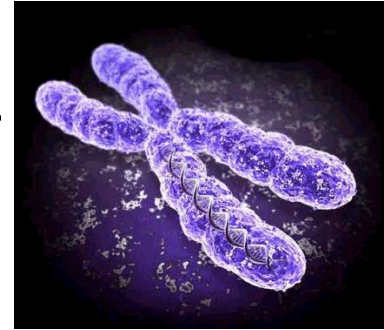
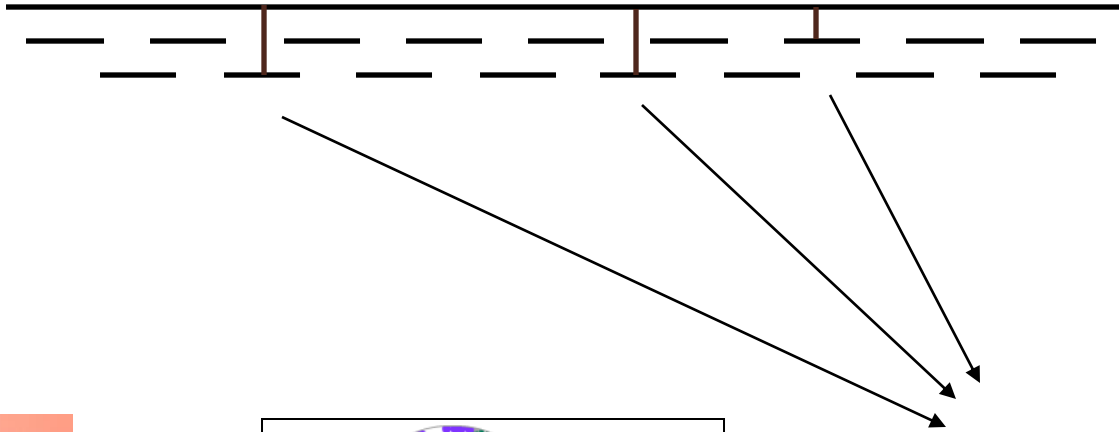
SCIENCE • VOL. 376, NO. 6588 • 01 MAR 2022

Read Mapping



Personal Genomics

How does your genome compare to the reference?



Heart Disease

Cancer

Amazing actress